

ECOG 315 / ECON 181, Summer 2025
Advanced Research Methods and Statistical Programming
Week 6 Lecture Slides

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Introduction to Dynamic Models

- ▶ Want to model decisions and outcomes of **agents** over (discrete) time
- ▶ Formal name: microeconomic dynamic stochastic optimization problems
- ▶ Outcomes probably subject to risk: random shocks from some distribution
- ▶ Standard assumptions: agents know their model and observe their current state, choose optimal action according to their preferences
- ▶ Almost always assume that agents have **time consistent preferences**: expectations of preferences in future align with actual preferences in future
- ▶ Upshot: want to maximize discounted sum of utility flows, geometric discounting

Ingredients of a MicroDSOP

What do we need to put into a MicroDSOP?

- ▶ Who are the agents? Who is this model about?
- ▶ How fast does time move? How long is a “period”?
- ▶ What do they want? What are their preferences?
- ▶ What do the agents observe about their situation when they act?
- ▶ What actions can agents take? What constraints do they face?
- ▶ What sources of risk/uncertainty to the agents face (shocks)?
- ▶ How do actions and shocks generate next period's state?

Jumping into the Deep End: Consumption-Saving Models

Let's sketch out a (very) basic consumption-saving model:

- ▶ Who is this model about? Person who earns income, consumes, and saves
- ▶ How long is a “period”? Usually a year or a quarter, depends
- ▶ What do they want? They like to consume; CRRA preferences (ρ)
- ▶ What do agents observe? They know their market resources M_t or cash-on-hand
- ▶ What actions can agents take? Divide M_t between consumption C_t and assets A_t
- ▶ What constraints do they face? Can't borrow assets: $A_t \geq 0$.
- ▶ What are the risks? Labor income Y_t is drawn iid from distribution F
- ▶ How does the situation change? Assets earn interest at risk-free factor $R = (1 + r)$

Translating the Model Into Math

consumer's problem: $\max_{C_t} \mathbb{E} \left[\sum_{t=0}^T \beta^t U(C_t) \mid M_0 \right]$ s.t.

$$U(C_t) = \frac{C_t^{1-\rho}}{1-\rho},$$

$$A_t = M_t - C_t,$$

$$A_t \geq 0,$$

$$K_{t+1} = A_t,$$

$$M_{t+1} = RK_{t+1} + Y_{t+1},$$

$$Y_{t+1} \sim F.$$

Translating the Model Into Math

consumer's (shorter) problem: $\max_{C_t} \mathbb{E} \left[\sum_{t=0}^T \beta^t \frac{C_t^{1-\rho}}{1-\rho} \mid M_0 \right]$ s.t.

$$C_t \leq M_t,$$

$$M_{t+1} = R(M_t - C_t) + Y_{t+1},$$

$$Y_{t+1} \sim F.$$

Taking it One Period at a Time: Bellman Representation

- ▶ The model has $T + 1$ periods, and T could be big
- ▶ T random income shocks will realize along the way
- ▶ Bellman insight: in period t , assume that you will act optimally in all later periods; now decision is only between current utility flow and future value (of resources)
- ▶ Bellman value function $V_t(X_t)$ maps from current state X_t to $\mathbb{R}$, representing expected PDV of utility flows from taking optimal actions from t onward
- ▶ Upshot: Can solve the model by backward induction, starting from $t = T$

Putting the Model into Bellman Form

$$V_0(M_0) = \max_{C_t} \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U(C_t) \mid M_0 \right] \text{ s.t.}$$

$$U(C_t) = \frac{C_t^{1-\rho}}{1-\rho},$$

$$C_t \leq M_t,$$

$$M_{t+1} = R(M_t - C_t) + Y_{t+1},$$

$$Y_{t+1} \sim F.$$

Putting the Model into Bellman Form

$$V_0(M_0) = \max_{C_t} \mathbb{E}_0 \left[U(C_0) + \sum_{t=1}^T \beta^t U(C_t) \mid M_0 \right] \text{ s.t.}$$

$$U(C_t) = \frac{C_t^{1-\rho}}{1-\rho},$$

$$C_t \leq M_t,$$

$$M_{t+1} = R(M_t - C_t) + Y_{t+1},$$

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$$V_0(M_0) = \max_{C_t} \left\{ U(C_0) + \mathbb{E}_0 \left[\sum_{t=1}^T \beta^t U(C_t) \mid M_0 \right] \right\} \quad \text{s.t.}$$

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Putting the Model into Bellman Form

$$V_0(M_0) = \max_{C_t} \left\{ U(C_0) + \beta \mathbb{E}_0 \left[\sum_{t=1}^T \beta^{t-1} U(C_t) \mid M_0 \right] \right\} \quad \text{s.t.}$$

$$U(C_t) = \frac{C_t^{1-\rho}}{1-\rho},$$

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Putting the Model into Bellman Form

$$V_0(M_0) = \max_{C_t} \left\{ U(C_0) + \beta \mathbb{E}_0 \mathbb{E}_1 \left[\sum_{t=1}^T \beta^{t-1} U(C_t) \mid M_1 \right] \right\} \quad \text{s.t.}$$

$$U(C_t) = \frac{C_t^{1-\rho}}{1-\rho},$$

$$C_t \leq M_t,$$

$$M_{t+1} = R(M_t - C_t) + Y_{t+1},$$

$$Y_{t+1} \sim F.$$

Putting the Model into Bellman Form

$$V_0(M_0) = \max_{C_t} \left\{ U(C_0) + \overbrace{\beta \mathbb{E}_0 \mathbb{E}_1 \left[\sum_{t=1}^T \beta^{t-1} U(C_t) \mid M_1 \right]}^{\equiv V_1(M_1)} \right\} \text{ s.t.}$$

$$U(C_t) = \frac{C_t^{1-\rho}}{1-\rho},$$

$$C_t \leq M_t,$$

$$M_{t+1} = R(M_t - C_t) + Y_{t+1},$$

$$Y_{t+1} \sim F.$$

Putting the Model into Bellman Form

$$V_0(M_0) = \max_{C_0} \{U(C_0) + \beta \mathbb{E}_0 V_1(M_1)\} \quad \text{s.t.}$$

$$U(C_t) = \frac{C_t^{1-\rho}}{1-\rho},$$

$$C_t \leq M_t,$$

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Putting the Model into Bellman Form

$$V_t(M_t) = \max_{C_t} \{U(C_t) + \beta \mathbb{E}_t V_{t+1}(M_{t+1})\} \quad \text{s.t.}$$

$$U(C_t) = \frac{C_t^{1-\rho}}{1-\rho},$$

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$$Y_{t+1} \sim F.$$

Solving By Backward Induction

- ▶ Bellman: the **value function** is all you need to know about the future
- ▶ Value function: expected PDV of making optimal choices from t onward (conditional on state)
- ▶ Start from the end of the problem, work backward
- ▶ Solving terminal period $t = T$ is trivial: there's no future, consume it all!
- ▶ Terminal period value function is then just $V_T(M_T) = U(M_T)$
- ▶ Solving non-terminal periods is where it gets more interesting

Solving the Terminal period

$$V_T(M_T) = \max_{C_T} \{U(C_T) + \beta \mathbb{E}_T V_{T+1}(M_{T+1})\} \quad \text{s.t.}$$

$$U(C_t) = \frac{C_t^{1-\rho}}{1-\rho},$$

$$C_t \leq M_t,$$

$$M_{t+1} = R(M_t - C_t) + Y_{t+1},$$

$$Y_{t+1} \sim F.$$

Solving the Terminal period

$$V_T(M_T) = \max_{C_T} \left\{ U(C_T) + \beta \mathbb{E}_T \overbrace{V_{T+1}(M_{T+1})}^{=0, \text{ no future}} \right\} \quad \text{s.t.}$$

$$U(C_t) = \frac{C_t^{1-\rho}}{1-\rho},$$

$$C_t \leq M_t,$$

$$M_{t+1} = R(M_t - C_t) + Y_{t+1},$$

$$Y_{t+1} \sim F.$$

Solving the Terminal period

$$V_T(M_T) = \max_{C_T} \{U(C_T) + \beta \cdot 0\} \quad \text{s.t.}$$

$$U(C_t) = \frac{C_t^{1-\rho}}{1-\rho},$$

$$C_t \leq M_t.$$

Solving the Terminal period

$$V_T(M_T) = \max_{C_T} \left\{ \frac{C_T^{1-\rho}}{1-\rho} + \right\} \quad \text{s.t.} \\ C_T \leq M_T.$$

Solving the Terminal period

$$V_T(M_T) = \frac{M_T^{1-\rho}}{1-\rho}.$$

Solving Non-Terminal Periods

- ▶ There is no closed form solution to non-terminal periods
- ▶ Problem can only be solved **numerically** on a computer
- ▶ Numeric solution: approximate representation of true solution to chosen accuracy
- ▶ Solve optimization problem on a finite set of state space gridpoints
- ▶ Represent value function and policy function (consumption function) with approximating interpolations
- ▶ But how do we actually do that? Several ways to go about this

Method 1: Explicit Value Maximization

- ▶ Bellman representation says to maximize sum of current period utility and expected discounted future value, so... let's do that
- ▶ Choose a set of gridpoints for M_t , starting at/near zero
- ▶ For each M_t , search for the C_t that maximizes utility plus expected future value
- ▶ Construct value function by interpolating on (M_t, V_t) pairs: connect the dots
- ▶ Construct consumption function by interpolating on (M_t, C_t) pairs

Method 2: First Order Conditions

- ▶ When we maximize $U(C_t) + \beta \mathbb{E} V_{t+1}(M_{t+1})$, we are implicitly finding a solution to the first order conditions for optimality
- ▶ Derivative of maximand w.r.t C_t : $U'(C_t) - R\beta \mathbb{E} V'_{t+1}(M_{t+1})$
- ▶ FOC for interior solution: that should be zero
- ▶ FOC for constrained solution: it should be positive, but $C_t = M_t$
- ▶ To do this, we need the **marginal value function** $V'_t(M_t)$
- ▶ Envelope condition: $V'_t(M_t) = U'(C_t)$ (for optimal C_t)
- ▶ Don't need to compute the value function at all! Consumption function is sufficient statistic for marginal value function!

Logic of the Envelope Condition

$$V_t(M_t) = \max_{C_t} \{U(C_t) + \beta \mathbb{E}_t V_{t+1}(M_{t+1})\},$$

$$C_t(M_t) \equiv \arg \max_{C_t} \{U(C_t) + \beta \mathbb{E}_t V_{t+1}(M_{t+1})\},$$

$$U'(C_t) = R\beta \mathbb{E}_t V'_{t+1}(M_{t+1}) \text{ (first order condition)}$$

Logic of the Envelope Condition

$$V_t(M_t) = U(\mathbf{C}_t(M_t)) + \beta \mathbb{E}_t V_{t+1}(M_{t+1}) \implies$$

$$V_t(M_t) = U(\mathbf{C}_t(M_t)) + \beta \mathbb{E}_t V_{t+1}(R(M_t - \mathbf{C}_t(M_t)) + Y_{t+1}) \implies$$

$$V'_t(M_t) = \mathbf{C}'_t(M_t)U'(\mathbf{C}_t(M_t)) + (1 - \mathbf{C}'_t(M_t))R\beta \mathbb{E}_t V'_{t+1}(R(M_t - \mathbf{C}_t(M_t)) + Y_{t+1}) \implies$$

$$V'_t(M_t) = \mathbf{C}'_t(M_t)U'(C_t) + (1 - \mathbf{C}'_t(M_t))R\beta \mathbb{E}_t V'_{t+1}(M_{t+1}) \implies$$

$$V'_t(M_t) = \mathbf{C}'_t(M_t)U'(C_t) + (1 - \mathbf{C}'_t(M_t))U'(C_t) \implies$$

$$V'_t(M_t) = (\mathbf{C}'_t(M_t) + 1 - \mathbf{C}'_t(M_t))U'(C_t) \implies$$

$$V'_t(M_t) = U'(C_t).$$

Method 3: Endogenous Grid Method

- ▶ Both prior methods involved **searching** for a solution at each gridpoint
- ▶ Make several “guesses” for optimal C_t at each M_t , compute expectations for each
- ▶ “I have this amount of M_t , how much should I consume in C_t ?”
- ▶ That’s a lot of “wasted” computation!
- ▶ EGM: Start at the end of the period with exogenous grid of A_t values
- ▶ Compute end-of-period marginal value of assets: with a tiny bit more A_t , how much more value do we expect to have in the future?
- ▶ Invert the FOC: what value of C_t must the agent have *just consumed* if ending the period at this A_t was the optimal choice?
- ▶ If you know A_t and C_t , then you also know M_t !

Rewriting the Problem for EGM

$$V_t(M_t) = \max_{C_t} \{U(C_t) + \beta \mathbb{E}_t V_{t+1}(M_{t+1})\} \quad \text{s.t.}$$

$$U(C_t) = \frac{C_t^{1-\rho}}{1-\rho},$$

$$C_t \leq M_t,$$

$$M_{t+1} = R(M_t - C_t) + Y_{t+1},$$

$$Y_{t+1} \sim F.$$

Rewriting the Problem for EGM

$$V_t(M_t) = \max_{C_t \leq M_t} \left\{ \frac{C_t^{1-\rho}}{1-\rho} + \beta \mathbb{E}_t V_{t+1}(\underbrace{R(M_t - C_t)}_{=A_t} + Y_{t+1}) \right\} \quad \text{s.t.} \quad Y_{t+1} \sim F.$$

Rewriting the Problem for EGM

$$V_t(M_t) = \max_{C_t \leq M_t} \left\{ \frac{C_t^{1-\rho}}{1-\rho} + \beta \mathbb{E}_t V_{t+1}(RA_t + Y_{t+1}) \right\} \quad \text{s.t.} \quad Y_{t+1} \sim F.$$

$$\mathfrak{V}_t(A_t) \equiv \beta \mathbb{E}_t V_{t+1}(RA_t + Y_{t+1}),$$

$$\mathfrak{V}'_t(A_t) = \beta R \mathbb{E}_t V'_{t+1}(RA_t + Y_{t+1}).$$

Rewriting the Problem for EGM

$$V_t(M_t) = \max_{C_t \leq M_t} \left\{ \frac{C_t^{1-\rho}}{1-\rho} + \mathfrak{V}_t(A_t) \right\} \quad \text{s.t.} \quad A_t = M_t - C_t.$$

$$\text{FOC: } C_t^{-\rho} - \mathfrak{V}'_t(A_t) = 0.$$

$$\text{soln: } C_t = \mathfrak{V}'_t(A_t)^{-1/\rho}, \quad M_t = A_t + C_t.$$